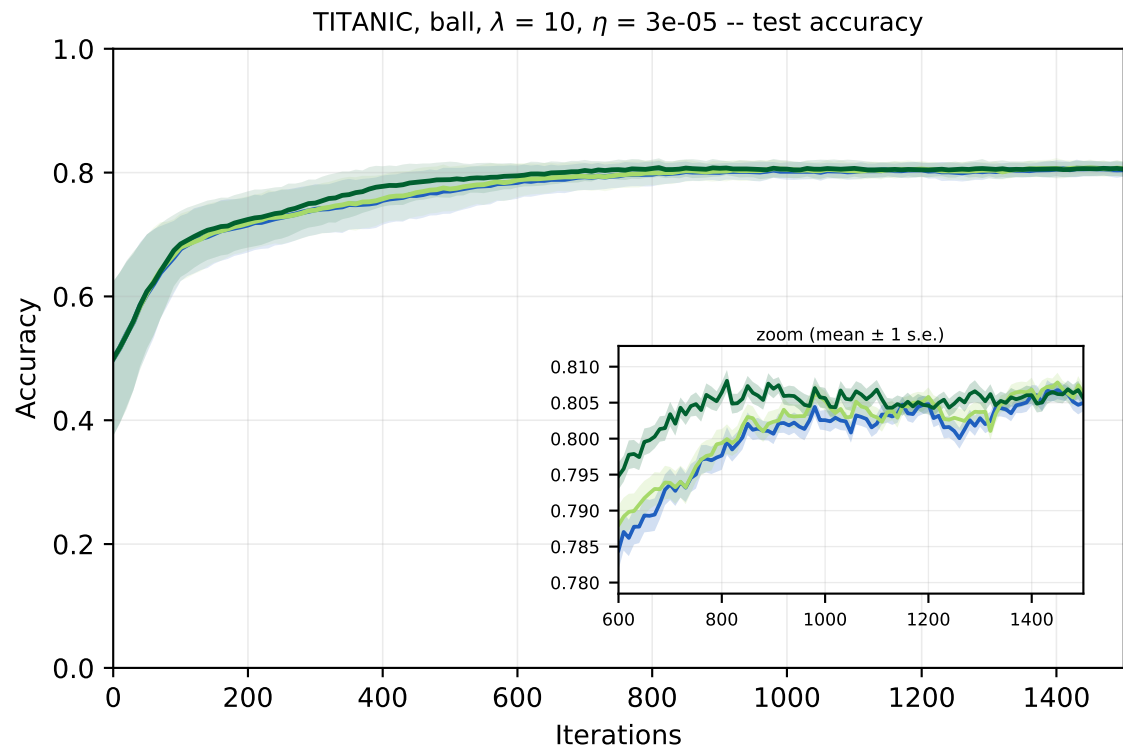
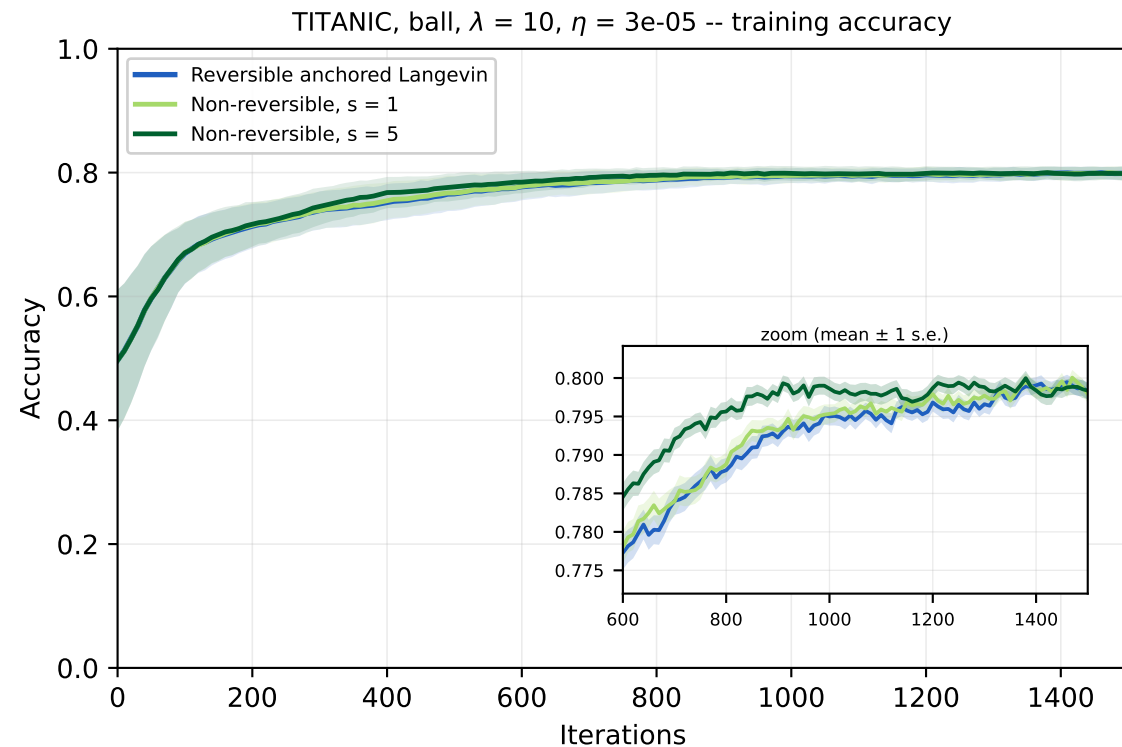


titanic_ball: $\lambda = 10$, $\eta = 3\text{e-}05$



TITANIC -- ball $\|w\|^2 \leq 2$ (radius $\sqrt{2}$). $d = 10$ (intercept w_0 + 9 features). $U = \sum \text{softplus} - y x \cdot w + w_0^2 / (2 \sigma^2) + \lambda \sum_{j=1}^9 |w_j|$; U_0 replaces $|w_j|$ by $\sqrt{w_j^2 + \delta^2}$; $a = \exp(U - U_0)$. EXACT gradient. $\sigma = 10$. $\lambda_{\text{lasso}} = 10$ with $\delta = \log 2 / (9 \lambda) = 0.007702$, so a stays in $[0.50, 1]$ at every λ (raising λ also sharpens the smoothing -- the two are coupled by that choice). Step $\eta = 3\text{e-}05$. $R = 100$ independent replicates on one fixed stratified 80/20 split ($n_{\text{train}} = 712$, $n_{\text{test}} = 179$). Bands: mean $\pm$ 1 SD (ddof = 1) -- repeat-run variability at a fixed split, not a confidence or credible interval. All λ are reported in `results/lambda/lambda_lasso.csv`.